CSCI-5622-001 Syllabus

CSCI-5622-001 Machine Learning

Spring 2026, 3 credit hours, 1/08/2026 - 4/24/2026

Mondays and Wednesdays 3:35pm-4:50pm Engineering Center ECCS 1B28

Final exam: 4/29/26 4:30pm-7pm

Instructor Information

Name: Esther Rolf
Office Location: ECES 122
My office hours are 2:30-3:30pm on Wednesdays, and by appointment (no office hours 1/19)
You can reach me at esther.rolf@colorado.edu
You can read more about my academic work and research on my website.

Course Staff

TAs

- **Nidhin Harilal (**Nidhin.Harilal@colorado.edu**).** Nidhin's office hours are:
  - Mondays 5-6pm in ECCS 114E
  - Fridays 11am-12pm in ECCS 114G
- **Ayoub Ghriss (**Ayoub.Ghriss@Colorado.EDU**).** Ayoub's office hours are:
  - Tuesdays 11am-12pm in ECCS 114E
  - Thursdays 2:30-3:30pm in ECCS 114F

Grader

- **Siva Tarun Reddy Ambala** (SivaTarunReddy.Ambala@colorado.edu)
  - For questions/regrade requests, email grader and cc professor Rolf

Piazza

This term we will be using Piazza for class discussion. The system is highly catered to getting you help fast and efficiently from classmates, the TA, and myself. Rather than emailing questions to the teaching staff, I encourage you to post your questions on Piazza.

Find our class signup link at: https://piazza.com/colorado/spring2026/csci5622001002

Course Information

Course description and purpose: This course trains students to build computer systems that learn from experience. It emphasizes practical and theoretical understanding of the most widely used algorithms (neural networks, decision trees, support vector machines) and covers connections to data mining and statistical modeling. It further trains students to address ethical considerations in their design and implementation of machine learning systems. *A strong foundation in probability, statistics, multivariate calculus, and linear algebra is highly recommended.*

Course prerequisites: CSCI 2820 or APPM 3310 or MATH 2130 or CSCI 3022 or APPM 4570 or APPM 3570 or STAT 4250 or MATH 3510 or CVEN 3227 or ECEN 3810 or ECON 3818 or MCEN 4120 (all min grade B). Restricted to Graduate Students Only.

Learning Outcomes

This course is structured to achieve the following learning outcomes:

- Gain a deep understanding of foundational ML algorithms spanning classification, regression, and unsupervised learning.
- Master the fundamentals of probabilistic prediction and the mathematics for representing data and modeling patterns in it.
- Obtain a good understanding of the core issues and challenges in machine learning, encompassing aspects such as data handling, model selection, model complexity.
- Design and implement various machine learning algorithms and develop insight into the advantages and limitations of popular machine learning methodologies.
- Understand how these foundational concepts contribute to modern deep learning
- Understand ethical implications of deploying machine learning algorithms in real-life.

Textbooks and Materials

Introduction to Machine Learning (4th Edition), Ethem Alpaydin, MIT Press, 2020, ISBN: 9780262043793, https://mitpress.mit.edu/9780262043793/introduction-to-machine-learning/
Learning from Data, Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin, 2012, https://amlbook.com/
The Elements of Statistical Learning (2nd Edition), Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer Series in Statistics, 2017, https://hastie.su.domains/ElemStatLearn/printings/ESLII_print12_toc.pdf.download.html
Additional reading material will be available on CANVAS.

Assignments

Homework

There will be a total of 6 homework assignments, comprising mathematical and programming problems. Homework assignments will extend the content seen in lecture, and provide you with a chance to engage deeper with the material on your own time. Truly understanding the homeworks will be an important component of preparing for the quizzes and final exam.

These assignments will be available on CANVAS. Each homework is worth 9 points. **Late submissions are accepted with a 2 day grace period after the deadline and 1 point penalty (i.e., for an 9 point assignment, 1 out of 9 points will be deducted per assignment).** After that, solutions will be posted and late assignments will no longer be accepted.

In class unit quizzes

There will be three quizzes in this class (15 points each). Quizzes will be held in class for the full duration of class on the following days:

- Wednesday Feb 11 (Covering unit 1: “algorithmic perspective to ML”)
- Wednesday March 11 (Covering unit 2: “data perspective to ML”)
- Wednesday April 15 (Covering unit 3: "probabilistic modeling” and unit 4: “optimization”)

While each quiz will focus on a specific unit, please note that the material in this class is cumulative, and thus content from prior units (or basic material from prerequisite classes) may be assessed as well.

Final exams

The final exam will be held on April 29th from 4:30-pm.

The final exam will be cumulative and cover all material presented during the class.

Grading Breakdown:

25% homework (6 homeworks x 5 points each, with lowest homework score dropped)

45% in class quizzes (15 points x 3 quizzes)

30% final exam (if your final exam score is higher than your lowest quiz score, replaces your lowest quiz score).

Regrade Policy

Regrade requests will be accepted only up to 5 business days after grades are released for each assignment/exam. Regrade requests should be sent to the course manager via email. Regrade requests sent after 5 business days after grades have been released will not be reviewed.

Course Calendar (2026)

A detailed version of the course calendar that will be updated throughout the semester with topics and assignment dates can be found here: 5622 calendar - Spr26.

Week 1 (1/12-1/14)
Introduction to ML, Algorithmic approaches to prediction, linear and logistic regression
Read syllabus and sign up for piazza/canvas

Week 2 (1/19-1/21) no class monday
linear and logistic regression, perceptron algorithm

Week 3 (1/26-1/28)
Classification algorithms: k-nearest neighbor, decision trees,

Week 4 (2/2-2/4)
random forests, boosted decision trees, data hygiene and evaluation

Week 5 (2/9-2/11)
Data preprocessing, **in-class quiz 2/11**

Week 6 (2/16-2/18)
Nonlinear regression, regularization, SVD

Week 7 (2/23-2/25)
Dimensionality reduction, PCA, embeddings, unsupervised learning

Week 8 (3/2-3/4)
Data generating models, multi-variate Gaussian distribution, clustering with k-means

Week 9 (3/9-3/11)
Guest lecture 3/9**, in-class quiz 3/11**

Week 9 (3/16-3/20) spring break; no class

Week 10 (3/23-3/25)
Probabilistic modeling, naive Bayes, probabilistic interpretations of linear and logistic regression

Week 11 (3/30-4/1)
Empirical risk minimization, loss functions, optimization methods

Week 11 (4/6-4/8)
Optimization methods (ct'd), intro to neural networks

Week 11 (4/13-4/15)
Intro to neural networks (ct'd), **in-class quiz 4/15**

Week 12 (4/20-4/22)
ML in practice, final review

Final exam: 4/29 4:30pm-7pm

Grading

The grading for the class will be as follows:

A	Numerical Grade 93 & Above
A-	Numerical Grade 90-92
B+	Numerical Grade 86-89
B	Numerical Grade 83-85
B-	Numerical Grade 80-82
C+	Numerical Grade 76-79
C	Numerical Grade 73-75
C-	Numerical Grade 70-72
D+	Numerical Grade 66-69
D	Numerical Grade 63-65
D-	Numerical Grade 60-62
F	Numerical Grade 59 or Below

Using Canvas and Other Technologies

All course material, including slides, readings, and assignments, will be uploaded on CANVAS. Grades for each assignment, quiz, and exam will be available on CANVAS. Course-related discussions will be on Piazza.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression,

veteran status, political affiliation, or political philosophy.

For more information, see the classroom behavior policy, the Student Code of Conduct, and the Office of Institutional Equity and Compliance.

Acceptable use of AI in this class

You may use generative AI tools to aid your understanding of the material, **but not to generate your homework**. The entirety of the work you submit must be student-generated with proper critical evaluation and original analysis or as outlined for each individual assignment. Course staff may conduct interview-based grading at any point during the semester, so you should always be prepared to explain what you have submitted. Exams and quizzes are closed books, so there should be no use of technology, including AI on exams.

Requirements for Infectious Disease

Members of the CU Boulder community and visitors to campus must follow university, department, and building health and safety requirements and all applicable campus policies and public health guidelines to reduce the risk of spreading infectious diseases. If public health conditions require, the university may also invoke related requirements for student conduct and disability accommodation that will apply to this class.

If you feel ill and think you might have COVID-19 or if you have tested positive for COVID-19, please stay home and follow the guidance of the Centers for Disease Control and Prevention (CDC) for isolation and testing. If you have been in close contact with someone who has COVID-19 but do not have any symptoms and have not tested positive for COVID-19, you do not need to stay home but should follow the guidance of the CDC for masking and testing.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the Disability Services website. Contact Disability Services at 303-492-8671 or dsinfo@colorado.edu for further assistance. If you have a temporary medical condition, see Temporary Medical Conditions on the Disability Services website.

If you have a required medical isolation for which you require adjustment, please notify the instructor 24 hours in advance when possible.

Preferred Student Names and Pronouns

CU Boulder recognizes that students' legal information doesn't always align with how they identify. Students may update their preferred names and pronouns via the student portal; those preferred names and pronouns are listed on instructors' class rosters. In the absence of such updates, the name that appears on the class roster is the student's legal name.

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: honor@colorado.edu, 303-492-5550. Students found responsible for violating the Honor Code will be assigned resolution outcomes from the Student Conduct & Conflict Resolution as well as be subject to academic sanctions from the faculty member. Visit Honor Code for more information on the academic integrity policy.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering an inclusive and welcoming learning, working, and living environment. University policy prohibits protected-class discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner violence (dating or domestic violence), stalking, and related retaliation by or against members of our community on- and off-campus. These behaviors harm individuals and our community. The Office of Institutional Equity and Compliance (OIEC) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email cureport@colorado.edu. Information about university policies, reporting options, and support resources can be found on the OIEC website.

Please know that faculty and graduate instructors must inform OIEC when they are made aware of incidents related to these policies regardless of when or where something occurred. This is to ensure that individuals impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support for a variety of concerns, visit the Don’t Ignore It page.

Religious Accommodations

Campus policy requires faculty to provide reasonable accommodations for students who, because of religious obligations, have conflicts with scheduled exams, assignments or required attendance. Please communicate the need for a religious accommodation in a timely manner. In this class, please notify the instructor at least 48 hours ahead of time.

See the campus policy regarding religious observances for full details.

Mental Health and Wellness

The University of Colorado Boulder is committed to the well-being of all students. If you are struggling with personal stressors, mental health or substance use concerns that are impacting academic or daily life, please contact Counseling and Psychiatric Services (CAPS) located in C4C or call (303) 492-2277, 24/7.

Free and unlimited telehealth is also available through Academic Live Care. The Academic Live Care site also provides information about additional wellness services on campus that are available to students.